

Constructive Ramsey Lower Bounds AND Chromatic Number of $\mathbb{R}^n$!

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Credit Where Credit is Due

This talk is based on

- 1) A great but hard-to-read paper by Frankl & Wilson.
- 2) A great exposition of it by Danesh Sivakumar.
(William Gasarch's master's student.)
- 3) A great set of slides by Kelin Zhu.
(An undergraduate working with William Gasarch.)

Easy But Weak Lower Bounds on $R(k)$

$$R(k) = \Omega(k^2)$$

Thm $R(k) > (k - 1)^2$.

Proof We color the edges of $K_{(k-1)^2}$ so there is no homog set of size k .

Here is our coloring:

Partition the vertices of $K_{(k-1)^2}$ into $k - 1$ blocks of $k - 1$ vertices.

- 1) Within each block color every edge RED.
- 2) Color every edge between blocks i and j BLUE.

Exercise Show that there is no homog set of size k .

End of Proof

Lemma Needed For $R(k) = \Omega(k^{\log_4 17})$

Lemma $R(a) > u$ and $R(b) > v$ implies $R(ab) > uv$.

Proof We first show that $(R(a) > u$ and $R(b) > v)$ implies $R(ab) > uv$.

Assume

- 1) $R(a) > u$ via a coloring COL_1 of the edges of K_u .
- 2) $R(b) > v$ via a coloring COL_2 of the edges of K_v .

We define a coloring of the edges of K_{uv} with no homog set of size ab .

Color the edges of K_{uv} as follows:

Partition the vertices of K_{uv} into u blocks of v vertices.

- 1) Within each block use COL_2 .
- 2) Color every edge between blocks i and j with $\text{COL}_1(i, j)$.

Exercise Show that there is no homog set of size ab .

End of Proof

$$R(k) = \Omega(k^{\log_4 17})$$

Thm $R(k) = \Omega(k^{\log_4 17})$. Hence $R(k) = \Omega(k^{2.0437})$.

Proof Since $R(4) > 17$ we obtain, from the Lemma,

For all $m \geq 1$, $R(4^m) > 17^m = (4^m)^{\log_4 17}$.

Exercise Show that $R(k) = \Omega(k^{\log_4 17})$.

End of Proof

Exercise Show that if $R(k_0) > m_0$ then $R(k) = \Omega(k^{\log_{k_0} m_0})$.

Exercise Use the last exercise to obtain a better lower bound than $R(k) = \Omega(k^{\log_4 17})$.

Two Combinatorial Lemmas

**We Will Prove Two
Combinatorial Lemmas**

Convention

We will often have sets $w \subseteq [n]$.

We will use w for both

- ▶ the set, and
- ▶ the bit vector.

We use linear algebra with those vectors over $\mathbb{F}_2$ or $\mathbb{F}_p$ or $\mathbb{R}$.

If I have $(\forall i \neq j)$ we mean all distinct indices.

If I have $w_1, \dots, w_m$ then these are all distinct.

Oddtown

All $\equiv$ are mod 2.

- 1) Oddtown has n people
- 2) They form clubs $w_1, \dots, w_m$ which are not necessarily disjoint.
- 3) Each club has an odd number of people. $(\forall i)[|w_i| \not\equiv 0]$.
- 4) Every two clubs have an even-sized intersection. So $(\forall i \neq j)[|w_i \cap w_j| \equiv 0]$.

Next slide is examples.

Examples of Oddtown

$$(\forall i)[|w_i| \equiv 1].$$

$$(\forall i \neq j)[|w_i \cap w_j| \equiv 0].$$

$$w_1 = \{1\}, \dots, w_n = \{n\}.$$

Hence $m = n$.

$$w_1 = \{1, 2, 3\}, w_2 = \{1, 2, 4\}, \dots, w_{n-2} = \{1, 2, n\}.$$

Hence $m = n - 2$.

Is it possible to have $m > n$?

Vote: Y, N, Unknown to Megan.

Answer on next slide.

Oddtown Theorem

Thm Let $n \in \mathbb{N}$. Let $w_1, \dots, w_m \subseteq [n]$ such that

$$(\forall i)[|w_i| \equiv 1].$$

$$(\forall i \neq j)[|w_i \cap w_j| \equiv 0].$$

Then $m \leq n$.

Proof View $w_1, \dots, w_m$ as vectors in $\mathbb{F}_2^n$. $\equiv$ is mod 2.

$$w_i \cdot w_i \equiv |w_i| \equiv 1.$$

$$w_i \cdot w_j \equiv |w_i \cap w_j| \equiv 0 \text{ for } i \neq j.$$

Rest of proof on next slide. (So write down theorem.)

Proof of Oddtown (cont)

We show that $w_1, \dots, w_m$ are linearly independent.

Assume $\alpha_1 w_1 + \dots + \alpha_m w_m \equiv 0$. We show $(\forall i)[\alpha_i \equiv 0]$.

DOT both sides by w_i .

$$\alpha_1(w_1 \cdot w_i) + \dots + \alpha_m(w_m \cdot w_i) \equiv 0.$$

For $i \neq j$, $w_i \cdot w_j \equiv 0$, so only the i th term remains.

$$\alpha_i(w_i \cdot w_i) \equiv 0.$$

Since $w_i \cdot w_i \equiv 1$:

$$\alpha_i \equiv 0.$$

So $(\forall i)[\alpha_i \equiv 0]$. Hence $w_1, \dots, w_m$ are linearly independent.

Since $w_1, \dots, w_m \in \mathbb{F}_2^n$ are linearly independent, $m \leq n$.

End of Proof

Fisher's Inequality (sorry, no cute name)

$n, v \in \mathbb{N}$, $\lambda \in \mathbb{Z}_{\geq 0}$, with $v \geq 1$.

- 1) Fishtown has n people
 - 2) They form clubs $w_1, \dots, w_m$ which are not necc disjoint.
 - 3) Every club has v people, where $v \geq 1$. So $(\forall i)[|w_i| = v]$.
 - 4) Every 2 clubs have intersection λ . So $(\forall i \neq j)[|w_i \cap w_j| = \lambda]$.
- Next slide is examples.

Examples of Fishtown

$$(\forall i)[|w_i| = v]$$
$$(\forall i \neq j)[|w_i \cap w_j| = \lambda].$$

$$v = 2, \lambda = 1$$

$$w_1 = \{1, n\}, w_2 = \{2, n\}, \dots, w_{n-1} = \{n-1, n\}. \text{ So } m = n-1.$$

Is it possible to have $m > n$?

Vote: Y, N, Unknown to Gary.

Answer on next slide.

Fisher's Inequality

Thm Let $n, v \in \mathbb{N}$ and $\lambda \in \mathbb{Z}_{\geq 0}$, with $v \geq 1$. Let $w_1, \dots, w_m \subseteq [n]$ such that

$$(\forall i)[|w_i| = v].$$

$$(\forall i \neq j)[|w_i \cap w_j| = \lambda].$$

Then $m \leq n$.

Proof If $m \leq 1$, the claim is immediate. Otherwise there are four cases.

Case 1 $\lambda = 0$. All w_i disjoint so $m \leq n$.

Case 2 $v < \lambda$ (assume $m \geq 2$): $\lambda = |w_i \cap w_j| \leq |w_i| = v < \lambda$. Cannot occur.

Case 3 $v = \lambda$ and $m \geq 2$ imply $w_i \cap w_j = w_i$ so $w_i \subseteq w_j$. By symmetry $w_j \subseteq w_i$. Hence $w_i = w_j$ contrary to convention.

Case 4 $1 \leq \lambda < v$. Next slide.

Proof of Fisher's Inequality Case 4

Case 4 $1 \leq \lambda < v$. Let $\mu + \lambda = v$.

We view $w_1, \dots, w_m$ as vectors in $\mathbb{R}^n$.

$$w_i \cdot w_i = v \quad w_i \cdot w_j = \lambda \text{ for } i \neq j.$$

We show that $w_1, \dots, w_m$ are linearly independent.

Assume $\alpha_1 w_1 + \dots + \alpha_m w_m = 0$. We show $(\forall i)[\alpha_i = 0]$.

DOT both sides by w_m .

$$\alpha_1(w_1 \cdot w_m) + \dots + \alpha_{m-1}(w_{m-1} \cdot w_m) + \alpha_m(w_m \cdot w_m) = 0.$$

$$w_1 \cdot w_m = \dots = w_{m-1} \cdot w_m = \lambda. \quad w_m \cdot w_m = v = \mu + \lambda.$$

$$\lambda \sum_{i=1}^{m-1} \alpha_i + (\mu + \lambda) \alpha_m = 0.$$

Add $\lambda \alpha_m$ to the first sum and subtract from the last term.

$$\lambda \sum_{i=1}^m \alpha_i + \mu \alpha_m = 0.$$

$$\alpha_m = \frac{-\lambda \sum_{i=1}^m \alpha_i}{\mu} \quad (\text{Used } \mu \neq 0).$$

Proof continued on next slide.

Finishing up the Proof of Fisher's Inequality

$$\alpha_m = \frac{-\lambda \sum_{i=1}^m \alpha_i}{\mu} \quad (\text{Used } \mu \neq 0).$$

What we did for α_m we could have done for any α_k .

$$\alpha_k = \frac{-\lambda \sum_{i=1}^m \alpha_i}{\mu}$$

Key All of the α_k 's are the same. Call them α .

$$\alpha_1 w_1 + \cdots + \alpha_m w_m = 0.$$

$$\alpha(w_1 + \cdots + w_m) = 0.$$

Each w_i is a 0-1 vector of weight $v \geq 1$. Therefore $w_1 + \cdots + w_m$ has nonnegative coordinates and coordinate sum $mv > 0$, so

$$w_1 + \cdots + w_m \neq 0.$$

Therefore $\alpha = 0$.

Hence every $\alpha_i = 0$, so $w_1, \dots, w_m$ are linearly independent.

Since they lie in $\mathbb{R}^n$, we obtain $m \leq n$.

End of Proof

Constructive Lower Bounds on $R(k)$

We Will Show

$$R(k) = \Omega(k^3)$$

$$R(k) = \Omega(k^3)$$

Thm $R(k) > \binom{k-1}{3} = \frac{k^3}{6} - k^2 + \frac{11k}{6} - 1$.

Proof We color the complete graph on $\binom{k-1}{3}$ vertices. Identify its vertices with $\binom{[k-1]}{3}$.

$$\text{COL}(x, y) = \begin{cases} \text{R} & \text{if } |x \cap y| \equiv 0 \pmod{2} \\ \text{B} & \text{if } |x \cap y| = 1 \end{cases} \quad (1)$$

Let $H = \{w_1, \dots, w_m\}$ be a homog set. Note that each w_i is a set. We show $m \leq k - 1$.

Case 1 H is **R**.

We show that $w_1, \dots, w_m$ satisfy the premise of Oddtown.

$$(\forall i)[|w_i| = 3 \equiv 1 \pmod{2}], \quad (\forall i \neq j)[|w_i \cap w_j| \equiv 0 \pmod{2}].$$

By Oddtown $m \leq k - 1$.

Case 2 on next slide.

$$R(k) = \Omega(k^3) \text{ (cont)}$$

Case 2 H is **B**.

We show that $w_1, \dots, w_m$ satisfy the premise of Fisher with $v = 3$ and $\lambda = 1$.

$$(\forall i)[|w_i| = 3], \quad (\forall i \neq j)[|w_i \cap w_j| = 1].$$

By Fisher's inequality $m \leq k - 1$.

End of Proof

Notes on the Proof that $R(k) = \Omega(k^3)$

1) The proof used **both** the Oddtown Theorem and Fisher's inequality.

2) The proof is constructive.

A constructive proof that $R(k) > f(k)$ is an algorithm that will, on input k , output a 2-colored graph on $f(k)$ vertices in time $p(f(k))$ where p is some polynomial.

Exercise Show that our proof that $R(k) > \binom{k-1}{3}$ is constructive.

3) There is a nonconstructive proof that shows $R(k) = \Omega(k2^{k/2})$.

Chromatic Number of $\mathbb{R}^n$

We Will Show

$$\chi(\mathbb{R}^n) = \Omega(n^2)$$

Chromatic Number of $\mathbb{R}^2$

Easy Thm For all 2-colorings of $\mathbb{R}^2$ there exist two points, an inch apart, that are the same color.

Slightly Harder Thm For all 3-colorings of $\mathbb{R}^2$ there exist two points, an inch apart, that are the same color.

Computer-Assisted Thm For all 4-colorings of $\mathbb{R}^2$ there exist two points, an inch apart, that are the same color.

Fairly Easy There is a 7-coloring of $\mathbb{R}^2$ where no two points, an inch apart are the same color.

Rephrasing Let $G = (V, E)$ be the graph $V = \mathbb{R}^2$ and $E = \{(x, y) : d(x, y) = 1\}$.

The above results can be summarized as $5 \leq \chi(G) \leq 7$.

We take a shortcut and write $5 \leq \chi(\mathbb{R}^2) \leq 7$.

Chromatic Number of $\mathbb{R}^n$

Let $n \in \mathbb{N}$. Let $G = (V, E)$ be the graph $V = \mathbb{R}^n$ and $E = \{(x, y) : d(x, y) = 1\}$.

We take a shortcut and call the graph $\mathbb{R}^n$.

What is $\chi(\mathbb{R}^n)$? The next slide gives some information.

Some Published Bounds on $\chi(\mathbb{R}^n)$

1) Upper and lower bounds on $\chi(\mathbb{R}^2)$, $\chi(\mathbb{R}^3)$, $\chi(\mathbb{R}^4)$.

$$5 \leq \chi(\mathbb{R}^2) \leq 7$$

$$6 \leq \chi(\mathbb{R}^3) \leq 15$$

$$9 \leq \chi(\mathbb{R}^4) \leq 49$$

2) Upper & lower bounds are known for all $\chi(\mathbb{R}^n)$ but are far apart.

3) Frankl-Wilson proved an exp lower bound. We will show some of the ideas later.

4) A standard asymptotic benchmark is

$$(1.239 + o(1))^n \leq \chi(\mathbb{R}^n) \leq (3 + o(1))^n.$$

Re-Look at Oddtown. $n \in \mathbb{N}$ throughout

Thm Let $w_1, \dots, w_m \subseteq [n]$ such that
 $(\forall i)[|w_i| \equiv 1]$ and $(\forall i \neq j)[|w_i \cap w_j| \equiv 0]$.
Then $m \leq n$.

Let's look at a special case.

Thm Let $w_1, \dots, w_m \subseteq [n]$ such that
 $(\forall i)[|w_i| = 3]$ and $(\forall i \neq j)[|w_i \cap w_j| \in \{0, 2\}]$.
Then $m \leq n$.

Take the contrapositive:

Thm Let $w_1, \dots, w_m \subseteq [n]$. Assume $(\forall i)[|w_i| = 3]$. If $m > n$ then
 $(\exists i \neq j)[|w_i \cap w_j| = 1]$.

Notation We call that last theorem **Oddtown'**.

Easy Algebra Lemma

We will need a value of c such that

$$\frac{\binom{n}{3}}{c} > n$$

$$\frac{n(n-1)(n-2)}{6} > nc$$

$$\frac{(n-1)(n-2)}{6} > c$$

We will use $c < \frac{(n-1)(n-2)}{6}$.

Lower Bound on $\chi(\mathbb{R}^n)$

Thm Let $c < \frac{(n-1)(n-2)}{6}$. $\forall \text{COL}: \mathbb{R}^n \rightarrow [c]$
 $\exists$ two points, same color, an inch apart.

Proof For $1 \leq i < j < k \leq n$ let w_{ijk} be

$$(0, \dots, 0, \frac{1}{2}, 0, \dots, 0, \frac{1}{2}, 0, \dots, 0, \frac{1}{2}, 0, \dots, 0).$$

The $\frac{1}{2}$'s are at the i th, j th, and k th spots.

We look at $W = \{w_{ijk} : 1 \leq i < j < k \leq n\}$.

$|W| = \binom{n}{3}$. By the pigeonhole principle, there is an $X \subseteq W$ such that

$|X| \geq \binom{n}{3}/c > n$, and

X is **monochromatic**.

Proof continued on next slide.

Lower Bound on $\chi(\mathbb{R}^n)$ Using X

$$W = \{w_{ijk} : 1 \leq i < j < k \leq n\}.$$

$$X \subseteq W, |X| > n.$$

We identify X with $S = \{\text{supp}(w) : w \in X\}$.

$$S \subseteq \binom{[n]}{3} \text{ and } |S| = |X| > n.$$

S satisfies premise of Oddtown', so $\exists x, y \in S, |x \cap y| = 1$.

x, y are 3-subsets with $|x \cap y| = 1$.

Let $x', y' \in X$ be the vectors whose supports are x and y .

Since $|x \cap y| = 1$, the supports differ in four coordinates.

Each coordinate where x', y' differ, one has a 0 and the other has a $\frac{1}{2}$. Hence

$$d(x', y')^2 = 4\left(\frac{1}{2}\right)^2 = 1$$

So $d(x', y') = 1$.

End of Proof

Note Only used Oddtown.

Asymptotic Lower Bound on $\chi(\mathbb{R}^n)$

Thm Let $c < \frac{(n-1)(n-2)}{6}$. $\forall \text{COL}: \mathbb{R}^n \rightarrow [c]$
 $\exists$ two points, same color, an inch apart.

Hence $\chi(\mathbb{R}^n) \geq \frac{(n-1)(n-2)}{6} = \Omega(n^2)$.

Want Better Results!

Used ODD and FISH to get
 $R(k) = \Omega(k^3)$.

Used ODD to get
 $\chi(\mathbb{R}^n) = \Omega(n^2)$.

Need better versions of ODD
& FISH to get improvements.

Generalized Oddtown: Elementary Bound

We use the elementary polynomial bound on a uniform family.

Theorem Let p be prime, $0 \leq v \leq n$, and $L \subseteq \{0, \dots, p-1\}$, with $v \bmod p \notin L$. Suppose $w_1, \dots, w_m \subseteq [n]$ are distinct and

$$|w_i| = v \quad (1 \leq i \leq m), \quad |w_i \cap w_j| \bmod p \in L \quad (i \neq j).$$

Then, writing $s = |L|$,

$$m \leq D(n, s) := \sum_{a=0}^s \binom{n}{a}.$$

Note We use $\binom{n}{a} = 0$ for $a > n$. If $0 \in L$, the constant monomial can be omitted, giving $m \leq D(n, s) - 1$. This recovers $m \leq n$ when $p = 2$, $L = \{0\}$, and v is odd.

For $s \leq v$, the chapter's Theorem A.6 proves the sharper bound $m \leq \binom{n}{s}$ by a rank argument over $\mathbb{Q}$.

The Polynomials and a Uniform Example

Work over $\mathbb{F}_p$. Identify w_i with its characteristic vector and set

$$f_i(x) = \prod_{\ell \in L} (x \cdot w_i - \ell).$$

For $i \neq j$, an intersection residue belongs to L , so $f_i(w_j) = 0$. On the diagonal,

$$f_i(w_i) = \prod_{\ell \in L} (v - \ell) \neq 0.$$

Example Take $p = 5$, $L = \{1, 3, 4\}$, $n = 8$, $v = 2$, and

$$w_1 = 11000000, \quad w_2 = 01100000.$$

Both sets have size 2, and their intersection has size 1. Thus

$$\begin{aligned} f_1(w_2) &= (1 - 1)(1 - 3)(1 - 4) = 0, \\ f_1(w_1) &= (2 - 1)(2 - 3)(2 - 4) = 2 \pmod{5}. \end{aligned}$$

Multilinearization and Dimension

Every f_i has degree $s = |L|$. On $\{0, 1\}^n$, replacing each x_j^a by x_j for $a \geq 1$ preserves all values and does not increase degree. Use these multilinear representatives from now on.

Their monomials are $\prod_{j \in A} x_j$ for $A \subseteq [n]$, $|A| \leq s$. Hence they lie in a vector space of dimension

$$D(n, s) = \sum_{a=0}^s \binom{n}{a}.$$

If $0 \in L$, every polynomial has zero constant term, reducing this dimension by one.

It remains to prove that the f_i are linearly independent.

Linear Independence Completes the Proof

Suppose $\sum_{i=1}^m \alpha_i f_i = 0$ as polynomials over $\mathbb{F}_p$. Evaluate this identity at w_j :

$$0 = \sum_{i=1}^m \alpha_i f_i(w_j) = \alpha_j f_j(w_j).$$

All off-diagonal terms vanish, whereas

$$f_j(w_j) = \prod_{\ell \in L} (v - \ell) \neq 0.$$

Since $\mathbb{F}_p$ is a field, $\alpha_j = 0$. This holds for every j .

The m polynomials are independent in a space of dimension $D(n, s)$, so

$$m \leq D(n, s).$$

This is exactly the same evaluation argument as before; uniformity simply makes the diagonal value independent of j .

Generalization of Fisher

Fisher Let $n, \lambda, v \in \mathbb{N}$. Let $w_1, \dots, w_m \subseteq [n]$ such that $(\forall i)[|w_i| = v]$ and $(\forall i \neq j)[|w_i \cap w_j| = \lambda]$.

Then $m \leq n$.

Generalized Fisher Let $n, v \in \mathbb{N}$. Let $L \subseteq \{0, \dots, v-1\}$. Let $w_1, \dots, w_m \subseteq [n]$ such that $(\forall i)[|w_i| = v]$ and $(\forall i \neq j)[|w_i \cap w_j| \in L]$.

Then $m \leq \sum_{a=0}^{|L|} \binom{n}{a}$.

Proof of the Generalization of Fisher

Generalized Fisher Let $n, v \in \mathbb{N}$. Let $L \subseteq \{0, 1, \dots, v-1\}$. Let $w_1, \dots, w_m \subseteq [n]$ such that $(\forall i)[|w_i| = v]$ and $(\forall i \neq j)[|w_i \cap w_j| \in L]$.
Then $m \leq \sum_{a=0}^{|L|} \binom{n}{a}$.

Proof

Let p be a prime, $p > v$.

We can apply Gen Oddtown to $w_1, \dots, w_m$ and L since $|w_i| = v$, $v \bmod p = v \notin L$, and $|w_i \cap w_j| \bmod p \in L$ for $i \neq j$.
 $|w_i \cap w_j|$ can't be v since then $w_i = w_j$.

By Gen Oddtown $m \leq \sum_{a=0}^{|L|} \binom{n}{a}$.

End of Proof

Better Lower Bound on $R(k)$

**We use Gen of ODD and
FISH to get better Lower
Bound on $R(k)$.**

A Bound We Will Need

Lemma Let $n, p \in \mathbb{N}$ such that $n \geq 4p$. Then $\sum_{a=0}^{p-1} \binom{n}{a} < 2\binom{n}{p-1}$.

Proof Use $1 \leq a \leq p-1$ on the first inequality, and $n \geq 4p$ on the second inequality to get:

$$\frac{\binom{n}{a-1}}{\binom{n}{a}} = \frac{a}{n-a+1} \leq \frac{p-1}{n-p+2} \leq \frac{p-1}{4p-p+2} = \frac{p-1}{3p+2} \leq \frac{1}{3}.$$

Hence $\binom{n}{a-1} \leq \frac{1}{3}\binom{n}{a}$, so $\binom{n}{a} \leq \frac{1}{3^{p-a-1}}\binom{n}{p-1}$

Therefore

$$\sum_{a=0}^{p-1} \binom{n}{a} < \binom{n}{p-1} \left(1 + \frac{1}{3} + \frac{1}{3^2} + \cdots \right) < 2\binom{n}{p-1}.$$

End of Proof

Lower Bound on $R(k)$

Thm Let p be a prime. Let $n > 2p^2$. Let $k = 2\binom{n}{p-1}$. Then $R(k) > \binom{n}{p^2-1}$.

Proof We color the complete graph on $\binom{n}{p^2-1}$ vertices. Identify its vertices with $\binom{[n]}{p^2-1}$.

$$\text{COL}(x, y) = \begin{cases} \mathbf{R} & \text{if } |x \cap y| \not\equiv p-1 \pmod{p} \\ \mathbf{B} & \text{if } |x \cap y| \equiv p-1 \pmod{p} \end{cases} \quad (2)$$

Let $H = \{w_1, \dots, w_m\}$ be a homog set. We show $m < k$.

Case 1

$$\text{COL}(x, y) = \begin{cases} \text{R} & \text{if } |x \cap y| \not\equiv p-1 \pmod{p} \\ \text{B} & \text{if } |x \cap y| \equiv p-1 \pmod{p} \end{cases} \quad (3)$$

Case 1 H is **R**. We show that H satisfies the premise of Generalized Oddtown.

$(\forall i)[|w_i| = p^2 - 1]$ and $(\forall i \neq j)[|w_i \cap w_j| \not\equiv p-1 \pmod{p}]$.

Let $L = \{0, \dots, p-2\}$.

$|w_i| = v = p^2 - 1$, $v \pmod{p} = p-1 \notin L$, and $|w_i \cap w_j| \pmod{p} \in L$ for $i \neq j$.

By Generalized Oddtown $m \leq \sum_{a=0}^{|L|} \binom{n}{a} = \sum_{a=0}^{p-1} \binom{n}{a}$.

By Lemma, and $n > 2p^2 \geq 4p$, we have $\sum_{a=0}^{p-1} \binom{n}{a} < 2\binom{n}{p-1} = k$.

Case 2

$$\text{COL}(x, y) = \begin{cases} \mathbf{R} & \text{if } |x \cap y| \not\equiv p-1 \pmod{p} \\ \mathbf{B} & \text{if } |x \cap y| \equiv p-1 \pmod{p} \end{cases} \quad (4)$$

Case 2 H is **B**. We show that $w_1, \dots, w_m$ satisfy the premise of Generalized Fisher.

$(\forall i)[|w_i| = p^2 - 1]$ and $(\forall i \neq j)[|w_i \cap w_j| \equiv p-1 \pmod{p}]$.

Let $v = p^2 - 1$ and $L = \{p-1, 2p-1, \dots, p(p-1)-1\}$.

$(\forall i)[|w_i| = p^2 - 1], \quad (\forall i \neq j)[|w_i \cap w_j| \in L]$.

By Generalized Fisher $m \leq \sum_{a=0}^{|L|} \binom{n}{a} = \sum_{a=0}^{p-1} \binom{n}{a}$.

By Lemma, and $n > 2p^2 \geq 4p$, we have $\sum_{a=0}^{p-1} \binom{n}{a} < 2\binom{n}{p-1} = k$.

End of Proof

Arbitrarily Large Polynomial Exponents

We have **Thm** Let p be a prime. Let $n > 2p^2$. Let $k = 2\binom{n}{p-1}$.

Then

$$R(k) > \binom{n}{p^2-1}.$$

Example Let $p = 101$. Then

$$\text{Let } k(n) = 2\binom{n}{101-1} = 2\binom{n}{100}.$$

$$R(k(n)) > \binom{n}{10200}.$$

Since $k(n) = \Theta(n^{100})$ and $\binom{n}{10200} = \Theta(n^{10200}) = \Theta(k(n)^{102})$ there is a constant $c > 0$ such that, for sufficiently large n

$$R(k(n)) \geq ck(n)^{102}.$$

Exercise Show that $R(k) = \Omega(k^{102})$.

Exercise For every fixed $d \in \mathbb{N}$, prove that $R(k)/k^d \rightarrow \infty$. Hint: choose a fixed prime $p > d$ and use an argument similar to ours.

The Result in the Frankl-Wilson Paper

We have **Thm** Let p be a prime. Let $n > 2p^2$. Let $k = 2\binom{n}{p-1}$.
Then

$$R(k) > \binom{n}{p^2-1}.$$

Using some number theory and approximations, and choosing p carefully, Frankl-Wilson obtained

$$R(k) \geq k^{(1+o(1))\frac{\log k}{4 \log(\log k)}}$$

Superpolynomial!

Better Lower Bound on $\chi(\mathbb{R}^n)$

**We use Gen of ODD to
get better Lower Bound
on $\chi(\mathbb{R}^n)$**

Uniform Generalized Oddtown: the Application

Let p be prime, $2p - 1 \leq n$, and let $w_1, \dots, w_m \subseteq [n]$ be distinct sets of size $v = 2p - 1$.

Take $L = \{0, \dots, p - 2\}$. Since $v \equiv p - 1 \pmod{p}$, the uniform theorem gives

$$(|w_i \cap w_j| \not\equiv p - 1 \pmod{p} \text{ for every } i \neq j) \implies m \leq D(n, p - 1).$$

By the contrapositive, if $m > D(n, p - 1)$, some distinct pair satisfies $|w_i \cap w_j| \equiv p - 1 \pmod{p}$.

Distinct $(2p - 1)$ -sets have intersection in $\{0, \dots, 2p - 2\}$. The only possible value with that residue is therefore

$$|w_i \cap w_j| = p - 1.$$

Notation Call this consequence Generalized Oddtown'.

Lower Bound on $\chi(\mathbb{R}^n)$

Thm Let $n, c \in \mathbb{N}$, let p be prime, and assume $2p - 1 \leq n$. If $c < \binom{n}{2p-1} / \sum_{a=0}^{p-1} \binom{n}{a}$, then $\forall \text{COL}: \mathbb{R}^n \rightarrow [c]$
 $\exists$ two points, same color, an inch apart.

Proof Let $\mathcal{A} = \binom{[n]}{2p-1}$. For $A \in \mathcal{A}$ define $v_A \in \mathbb{R}^n$ by

$$(v_A)_i = \begin{cases} 1/\sqrt{2p} & \text{if } i \in A \\ 0 & \text{if } i \notin A \end{cases} \quad (5)$$

Let $\mathcal{V} = \{v_A : A \in \mathcal{A}\}$. By the pigeonhole principle there is a monochromatic set $X \subseteq \mathcal{V}$ with

$$|X| \geq \frac{\binom{n}{2p-1}}{c} > \sum_{a=0}^{p-1} \binom{n}{a}.$$

Proof continues on next slide.

Lower Bound on $\chi(\mathbb{R}^n)$ (cont)

Recap

$$\mathcal{A} = \binom{[n]}{2p-1}.$$

$$\mathcal{V} = \{v_A : A \in \mathcal{A}\}.$$

$$|X| \geq \frac{\binom{n}{2p-1}}{c} > \sum_{a=0}^{p-1} \binom{n}{a}, X \text{ is monochromatic.}$$

New Material

$$\text{Let } \mathcal{S} = \{A \in \mathcal{A} : v_A \in X\}.$$

Then $|\mathcal{S}| = |X|$ and every member of $\mathcal{S}$ has size $2p - 1$.

Generalized Oddtown' gives distinct $A, B \in \mathcal{S}$ with $|A \cap B| = p - 1$.

Hence $|A \Delta B| = 2p$. On each coordinate where v_A and v_B differ, one vector has entry 0 and the other has entry $\frac{1}{\sqrt{2p}}$. Therefore

$$d(v_A, v_B)^2 = 2p \left(\frac{1}{\sqrt{2p}}\right)^2 = 1. \text{ Hence } d(v_A, v_B) = 1.$$

End of Proof

Note Only used Gen. Oddtown.

Asymptotic Lower Bound on $\chi(\mathbb{R}^n)$

Thm Let $n, c \in \mathbb{N}$ and let p be a prime such that $n \geq 4p$. If $c < \binom{n}{2p-1} / \sum_{a=0}^{p-1} \binom{n}{a}$, then $\forall \text{COL}: \mathbb{R}^n \rightarrow [c] \exists$ two points, same color, an inch apart.

By Lemma and $n \geq 4p$ we have $\sum_{a=0}^{p-1} \binom{n}{a} \leq 2\binom{n}{p-1}$. Hence

$$\chi(\mathbb{R}^n) \geq \frac{1}{2} \frac{\binom{n}{2p-1}}{\binom{n}{p-1}}.$$

$$\chi(\mathbb{R}^n) \geq \frac{(p-1)!}{2^{p+1}(2p-1)!} n^p.$$

End of Proof

Exercise For every fixed d , $\chi(\mathbb{R}^n) = \Omega(n^d)$. The constant in the Ω will depend on d .

Better is known. See next slide.

Prime Numbers Already Give an Exponential Bound

From our existing construction, whenever p is prime and $2p - 1 \leq n$,

$$\chi(\mathbb{R}^n) \geq \frac{\binom{n}{2p-1}}{D(n, p-1)}, \quad D(n, p-1) = \sum_{a=0}^{p-1} \binom{n}{a}.$$

We now allow p to grow with n , rather than fixing p .

Theorem With $\beta = (1 + \sqrt{2})/2 = 1.207106\dots$,

$$\chi(\mathbb{R}^n) \geq \exp((\log \beta + o(1))n).$$

Only primes, uniform generalized Oddtown, and the prime number theorem are needed. All logarithms on the following slides are natural.

A Prime Near the Desired Proportion

Fix $0 < \alpha < 1/4$. The prime number theorem implies that one can choose primes

$$p = p(n) = (\alpha + o(1))n.$$

Indeed, for every fixed $0 < \varepsilon < 1$,

$$\pi(x) - \pi((1 - \varepsilon)x) \sim \frac{\varepsilon x}{\log x} > 0.$$

Thus the largest prime at most x has ratio to x tending to 1. Apply this with $x = \alpha n$.

In particular $p < n/4$ for all sufficiently large n . The earlier binomial-sum lemma gives

$$D(n, p - 1) < 2 \binom{n}{p - 1}.$$

No assumption on measurable color classes is involved.

Estimating the Binomial Ratio

Let $h(t) = -t \log t - (1 - t) \log(1 - t)$ for $0 < t < 1$. Stirling's formula gives, for $r/n \rightarrow t \in (0, 1)$,

$$\log \binom{n}{r} = nh(r/n) + O(\log n) = nh(t) + o(n).$$

For $p = (\alpha + o(1))n$, the finite construction therefore yields

$$\begin{aligned} \log \chi(\mathbb{R}^n) &\geq \log \binom{n}{2p-1} - \log \binom{n}{p-1} - \log 2 \\ &= n(h(2\alpha) - h(\alpha)) + o(n). \end{aligned}$$

The factor 2 and the shifts by 1 contribute only $o(n)$ to the logarithm. We must maximize $g(\alpha) = h(2\alpha) - h(\alpha)$.

Optimizing the Exponential Base

On the admissible interval $0 < \alpha < 1/4$,

$$g'(\alpha) = 2 \log \frac{1-2\alpha}{2\alpha} - \log \frac{1-\alpha}{\alpha} = \log \frac{(1-2\alpha)^2}{4\alpha(1-\alpha)}.$$

The derivative changes from positive to negative at

$$\alpha_* = \frac{2 - \sqrt{2}}{4} = 0.146446\dots < \frac{1}{4}.$$

Substitution gives

$$g(\alpha_*) = \log \frac{1 + \sqrt{2}}{2} = \log \beta.$$

Hence $\log \chi(\mathbb{R}^n) \geq (\log \beta + o(1))n$. This optimizes the exponential rate within this particular binomial-ratio construction.

Conclusion: Primes Already Suffice

Immediate goal: The primes case alone gives exponential growth with base $\beta = 1.207106\dots$. For every $1 < b < \beta$,

$$\chi(\mathbb{R}^n) \geq b^n \quad \text{for all sufficiently large } n.$$

Taking $b = 1.2$ proves the lower bound requested in the outline.

It also implies $\chi(\mathbb{R}^n)/n^d \rightarrow \infty$ for every fixed d .

Precision: A statement $\beta^{n+o(n)}$ describes an exponential rate. It does not by itself assert a multiplicative $(1 + o(1))\beta^n$ factor. The slack between 1.2 and β makes the outline's weaker base harmless.

Prime powers are unnecessary for this conclusion; an easy uniform prime-power extension is included in the appendix.

Ramsey Numbers: Filling in the Missing Interpolation

The same edge rule actually gives the slightly sharper finite statement

$$R(D_p + 1) > N_p, \quad D_p = \sum_{j=0}^{p-1} \binom{p^3}{j}, \quad N_p = \binom{p^3}{p^2 - 1}.$$

Both red and blue homogeneous families have size at most D_p by the uniform lemmas already proved. The modulus is still the prime p ; the set size $p^2 - 1$ is not a modulus.

We will show that, for every $\varepsilon > 0$ and all sufficiently large k ,

$$R(k) > \exp\left((1 - \varepsilon) \frac{(\log k)^2}{4 \log \log k}\right).$$

The construction and the proof use primes alone.

Two Elementary Estimates

For $1 \leq t \leq M$,

$$(M/t)^t \leq \binom{M}{t} \leq (eM/t)^t.$$

The lower bound follows by multiplying $(M-i)/(t-i) \geq M/t$; the upper bound follows from $t! \geq (t/e)^t$. Since $\binom{p^3}{j}$ increases for $0 \leq j \leq p-1$,

$$\binom{p^3}{p-1} \leq D_p \leq p \binom{p^3}{p-1}.$$

Consequently

$$\log D_p = 2p \log p + O(p),$$

$$\log N_p = p^2 \log p + O(p^2).$$

These estimates require no information about the distribution of primes.

Passing from Prime Thresholds to Every Large k

Choose the largest prime p with $D_p < k$, and let q be the next prime. The sequence D_p strictly increases and is unbounded. Thus

$$D_p < k \leq D_q.$$

The prime number theorem gives $q/p \rightarrow 1$. The preceding estimates then squeeze

$$\log k = (2 + o(1))p \log p, \quad \log \log k = (1 + o(1)) \log p.$$

In particular

$$p = (1 + o(1)) \frac{\log k}{2 \log \log k}.$$

Use the graph on $N(k) = N_p$ vertices: it has no homogeneous k -set, and

$$\log N(k) = (1 + o(1)) \frac{(\log k)^2}{4 \log \log k}.$$

Two Corrections to the Ramsey Writeup

Li-Lin, Theorem 6.4, printed p. 137. In the book's notation, replace $4 \log \log t$ by $2 \log \log t$ in both displayed bounds on p : $\log t \sim 2p \log p$ gives $p \sim \log t / (2 \log \log t)$.

The printed scale would instead give

$$\log \binom{p^3}{p^2 - 1} \sim p^2 \log p \sim \frac{(\log t)^2}{16 \log \log t} = \frac{1}{4} \omega(t),$$

inconsistent with the proof's claimed $\exp((1 - \varepsilon)\omega(t))$ vertex count for small ε . The theorem itself is correct.

Outline item 10.5 (using the outline's k). The asymptotic equality describes the construction size $N(k)$. For $R(k)$, write the lower bound

$$R(k) \geq \exp \left((1 - o(1)) \frac{(\log k)^2}{4 \log \log k} \right).$$

Equality at this scale would contradict the probabilistic exponential lower bound.

Li-Lin, *Elementary Methods of Graph Ramsey Theory*, p. 137.

The Fixed-Prime Exercise, for Every k

Fix a prime p and put $r = p^2 - 1$. Given a large integer k , set

$$M = \left\lfloor \left(\frac{k}{2p} \right)^{1/(p-1)} \right\rfloor.$$

Then

$$\sum_{j=0}^{p-1} \binom{M}{j} \leq pM^{p-1} \leq k/2 < k.$$

So the same coloring on $\binom{M}{r}$ vertices has no homogeneous k -set. For fixed p , $M = \Theta_p(k^{1/(p-1)})$, whence

$$\binom{M}{r} = \Theta_p(k^{r/(p-1)}) = \Theta_p(k^{p+1}).$$

Thus $R(k) = \Omega_p(k^{p+1})$ for every large k , not just a subsequence. Choose p with $p+1 > d$ to obtain $R(k)/k^d \rightarrow \infty$.

Appendix: An Easy Prime-Power Extension

Uniform forbidden-congruence theorem (Frankl-Wilson). Let $q = p^a$ with p prime and $a \geq 1$, and suppose $q - 1 \leq v \leq n$. If distinct sets $w_1, \dots, w_m \subseteq [n]$ satisfy

$$|w_i| = v, \quad |w_i \cap w_j| \not\equiv v \pmod{q} \quad (i \neq j),$$

then

$$m \leq \binom{n}{q-1}.$$

We add a binomial congruence and a rank argument over $\mathbb{Q}$. At $q = p$, this sharpens the bound used earlier in the Euclidean application; we kept the elementary polynomial proof in the main exposition.

This is the uniform prime-power result needed for our Euclidean construction. It does not assert an arbitrary- L extension modulo p^a with the bound $\binom{n}{|L|}$.

The One Lucas Congruence We Need

For every nonnegative integer N , write $N = dq + r$, $0 \leq r < q$. In polynomials modulo p ,

$$(1 + z)^N \equiv (1 + z^q)^d (1 + z)^r.$$

This follows by repeatedly applying $(1 + z)^p \equiv 1 + z^p \pmod{p}$. Comparing the coefficient of z^{q-1} gives

$$\binom{N}{q-1} \equiv \begin{cases} 1 & N \equiv q-1 \pmod{q}, \\ 0 & \text{otherwise} \end{cases} \pmod{p}.$$

Choose $0 \leq c < q$ with $v + c \equiv q-1 \pmod{q}$, and put

$$f(t) = \binom{t+c}{q-1}.$$

Then $f(v) \equiv 1$ and $f(|w_i \cap w_j|) \equiv 0$ for $i \neq j$.

Integer Values First, Then Reduction Modulo p

The polynomial $f(t) = \binom{t+c}{q-1}$ has rational coefficients, degree $s = q - 1$, and integer values for all nonnegative integer t . Form the integer matrix

$$A_{ij} = f(|w_i \cap w_j|).$$

The previous slide shows $A \equiv I \pmod{p}$. Therefore its determinant is nonzero modulo p and hence nonzero over the rationals:

$$\text{rank}_{\mathbb{Q}} A = m.$$

Each row comes from evaluating $f(\sum_{a \in w_i} x_a)$ at the incidence vectors of the v -sets w_j . Multilinearization preserves the values and gives degree at most s .

We next bound the dimension of these functions on the uniform layer.

Uniformity Reduces the Dimension to One Binomial

Write $x_S = \prod_{a \in S} x_a$. If $|S| = j \leq s \leq v$, then on incidence vectors of v -sets,

$$\sum_{\substack{T \supseteq S \\ |T|=s}} x_T = \binom{v-j}{s-j} x_S.$$

When S is contained in the set, both sides count its s -subsets containing S ; otherwise both sides are zero.

The binomial coefficient is positive. Dividing over $\mathbb{Q}$ expresses every monomial of degree at most s using the $\binom{n}{s}$ monomials of degree exactly s . Thus

$$m = \text{rank}_{\mathbb{Q}} A \leq \binom{n}{s} = \binom{n}{q-1}.$$

The division is over $\mathbb{Q}$, where it is legal; that coefficient may vanish modulo p .

Prime Powers: Application and Limitation

Assume $n \geq 2q - 1$ and take $v = 2q - 1$. A distinct intersection lies in $\{0, \dots, 2q - 2\}$, so its only possible value congruent to $v \pmod{q}$ is $q - 1$. Thus a family avoiding intersection size $q - 1$ has at most $\binom{n}{q-1}$ sets. Scaling incidence vectors by $1/\sqrt{2q}$ gives

$$\chi(\mathbb{R}^n) \geq \frac{\binom{n}{2q-1}}{\binom{n}{q-1}}.$$

Why a literal modulus substitution fails. The field with p^a elements has characteristic p , so incidence-vector dot products record counts only modulo p . For $a > 1$, the ring modulo p^a has zero divisors (e.g. $2 \cdot 2 = 0 \pmod{4}$).

In our proof, binomial coefficients are evaluated as integers before reduction modulo p . We never divide by $(q - 1)!$ in characteristic p . Frankl-Wilson (1981), Theorem 2, p. 358; proof, pp. 361-362.

Fisher's Inequality and Block Designs

Fisher's 1940 paper concerned incomplete block designs. Consider a nontrivial 2 -(v_0, k_0, λ) design with b blocks, $1 < k_0 < v_0$, and $\lambda > 0$. Every point lies in r blocks. Counting point-block incidences through a fixed point gives

$$r(k_0 - 1) = \lambda(v_0 - 1), \quad r > \lambda.$$

For each point x , let $A_x \subseteq [b]$ be the blocks containing x . These v_0 sets have size r , and any two have intersection λ . Our Fisher inequality applied to these sets gives

$$v_0 \leq b.$$

Source: R. A. Fisher (1940), *An examination of the different possible solutions of a problem in incomplete blocks*, *Annals of Eugenics* **10**, 52-75. Original paper.

Oddtown: Origin and the Useful Reference Trail

The problem originated with Erdős and was solved independently by Berlekamp (1969) and Graver (1975); the latter paper is titled *Boolean designs and self-dual matroids*. The name “Oddtown” comes from Babai and Frankl’s book, which presents the problem through a town’s residents and their clubs.

Bukh-Chao-Zheng (2025; revised August 2026) points to these sources:

- ▶ Babai-Frankl (1980), *On set intersections*, Theorem 1 and its proof: the prime-power divisibility argument. Bukh-Chao-Zheng cites this paper for prime-power Oddtown.
- ▶ Babai-Frankl, *Linear Algebra Methods in Combinatorics*: a textbook treatment of the linear-algebra method and its extensions.

The composite-modulus paper studies a different, broader problem; its new Fourier-analytic bounds are not needed for our prime-only proof. Bukh-Chao-Zheng, v2, introduction and references; Babai-Frankl (1980), pp. 103-105; Babai-Frankl, author’s book copy.